

Lecture 8

2/6/26
JRA

Homework Hints

• Problem 4.3 Resolving Power - Pointillism

Use blurring to cause dots in Pointillist painting to blur & look continuous

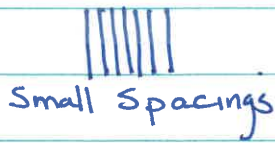
Here, if diffraction occurs to blur dots for the least likely to blur color, then diffraction will occur to blur all colors (i.e., if blue blurs, then all colors (blue, green, red) will blur).

Aside: Resolution is $\frac{1}{2}$ as good along direction of propagation as it is along direction perpendicular to direction of propagation (i.e., it is $\frac{1}{2}$ as good axially as it is laterally)

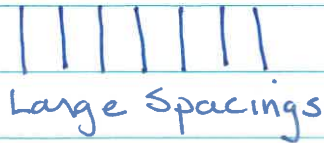
Abbe Theory of Image Formation (continued)

Abbe studied image formation by diffracting objects

- (1) Abbe used diffraction gratings w/ different line spacings, d , as models for objects w/ different spacings and sizes



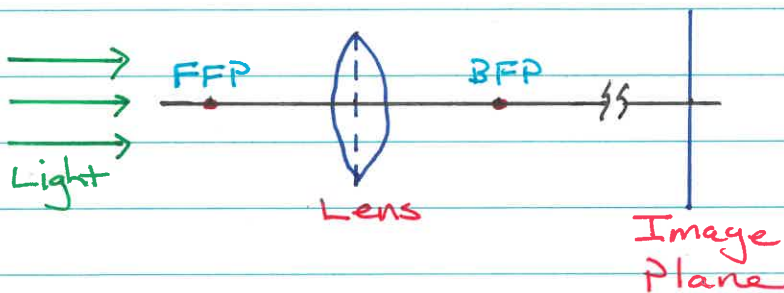
Model for small objects & "fine" details



Model for large objects & "coarse" details

(2) Abbe put gratings under μ scope

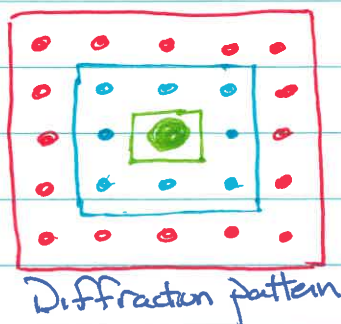
He observed an "image" in the image plane & a diffraction pattern in the back focal plane (BFP) of the objective.



FFP = front focal plane
 BFP = back focal plane
 ↓
 Back = opposite to incident light direction

(3) Abbe determined that an image w/ resolved object details only occurs if at least one higher diffraction order (e.g., $m = \pm 1$) is captured by the objective & appears in the BFP

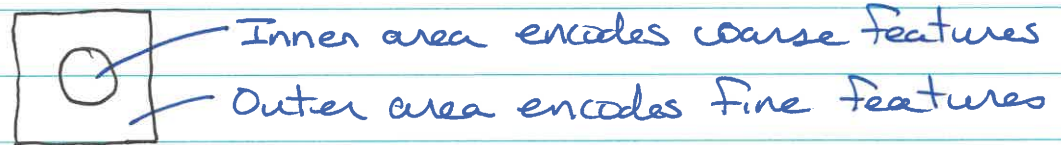
Passing only $m = 0$ gives an unresolved image



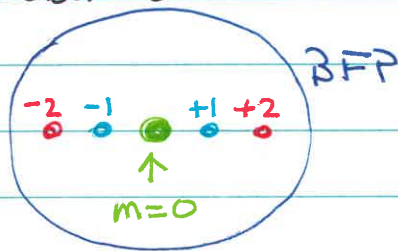
- Central spot ($m=0$) represents undiffracted light
- More central spots (e.g., blue) encode coarser details (which cause smaller-angle diffraction)
- More peripheral spots (e.g., red) encode fine details (which cause higher-angle diffraction)

MRI analog - MRI instrument collects diffraction pattern (k-space data) directly

Manipulating MRI diffraction pattern by masking features showed the following:



- (4) Abbe determined that capturing more (additional) diffraction spots improves resolution



- (5) Abbe derived an expression for the smallest resolvable spacing, d_{min}

$$d_{min} = \frac{0.5 \lambda_0}{NA}$$

Abbe

$$d_{min} = \frac{0.61 \lambda_0}{NA}$$

Rayleigh

Goal, to improve resolution, is to get a small d_{min}

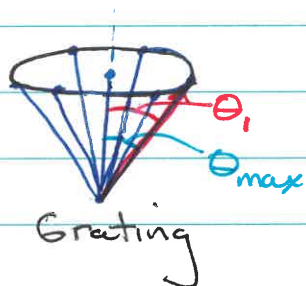
d_{min} gets smaller, $\therefore$ resolution improves, as $\begin{cases} \lambda_0 \downarrow \\ NA \uparrow \end{cases}$

(6) Filtering the diffraction pattern alters (determines) the image.

In MRI, filtering corresponds to how much of K space is collected, w/ limits

Too much MRI data $\Rightarrow$ Storage issues
 $\Rightarrow$ Noise issues

(*) Derivation of Abbe Equation (to w/in a factor of 2)



Diffraction spots are located at angles given by the expression

$$m\lambda = d \sin \Theta$$

As "d" $\downarrow$ angles $\uparrow$.

At some point, "d" is so small that the objective just captures Θ_1 (the first non-zero spot).

This is d_{min} when $\Theta_1 = \Theta_{max}$.

Evaluate Abbe expression for $m=1$ & $\Theta = \Theta_1 = \Theta_{max}$

$$m\lambda = d \sin \Theta \Rightarrow 1 \cdot \lambda = \lambda_0/n = d_{min} \sin \Theta_{max}$$

λ_0/n where "n" is the refractive index of the medium between the sample & lens

$$d_{\min} = \frac{\lambda_0}{n \sin \theta_{\max}} = \frac{\lambda_0}{NA}$$

The actual expression is $d_{\min} = \frac{\lambda_0}{2NA}$

The Factor of two improvement arises because you only need to capture one of the two first-order diffraction spots to get an image (i.e., $m = -1$ or $m = +1$ NOT both).

ACTIVITY 3

Abbe Theory

(See separate sheet for questions & answers.)

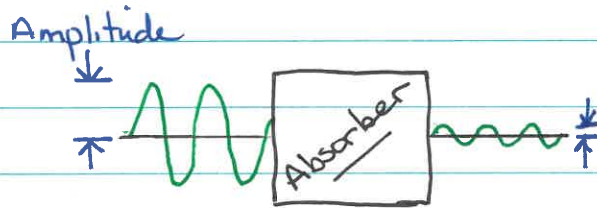
Contrast

Resolution is NOT enough for quality images... need contrast.

Many biological samples are essentially transparent. Need contrast to see features.

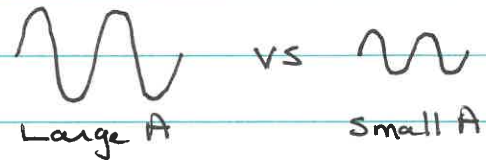
Contrast is $\left\{ \begin{array}{l} \text{Grayscale variation } \left\{ \begin{array}{l} \text{Dark/Bright} \\ \text{Black/Gray/White} \end{array} \right. \\ \text{Color variation } (\text{Blue/Green/Red}) \end{array} \right.$

Eye detects amplitude variations & color variations

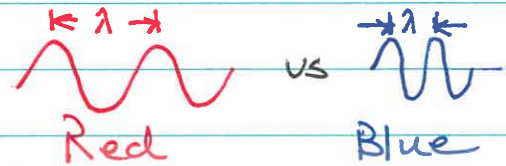


Absorbing amplitude specimen

Eye can distinguish amplitude variations

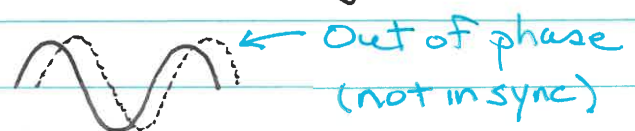
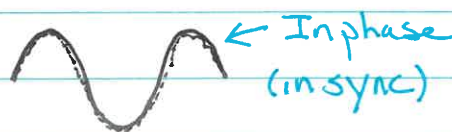


Eye can distinguish color variations



Cells don't absorb much, & they're not typically colored (except, e.g., cells w/ chlorophyll), so they are poor amplitude & color specimens

Cells do alter the "phase" of the light wave



Out of phase means the peaks & troughs of the waves do NOT superimpose

Eye cannot distinguish
phase variations
(phase shifts)



Biological specimens therefore tend to have very poor contrast unless you play "tricks" to enhance contrast

There are three prominent contrast-enhancing tools that do NOT involve altering the specimen:

- (1) Spatial filtering in the BFP of the objective
(where the diffraction pattern appears)
- (2) Altering illumination (e.g., darkfield microscopy)
- (3) Using interference (e.g., DIC)

It's also possible to alter the specimen

- (1) Staining w/ grayscale or color stains
- (2) Fluorescence

We will look at all of these approaches over the next few weeks

Physics 390 In-class Activity 3

Abbe Theory

Name: JRA

*First diffraction orders, Undiffracted light, Wavelength, Back focal plane,
Image plane, Numerical aperture, Horizontal lines*

Questions – choose the best answer from the choices above

1. The image appears in the Image Plane ?
2. Resolution improves with increasing numerical aperture ?
3. No image is generated if only undiffracted light is captured?
4. The diffraction pattern appears in the BFP ?
5. An image is formed if at least one of the first diffraction orders is captured?
6. Resolution improves with decreasing wavelength ?
7. Filtering to create a diffraction pattern consisting of central horizontal spots creates an image consisting of vertical lines?

↑ TRICK QUESTION
(w/ respect to provided answers!)

Selected Blackboard Images

Lecture 8

2/6/26

Abbe Theory of image formation by diffracting specimens

- 1) Diffracting gratings with different line spacings, d , used as models for objects with different spacings and sizes.

|||||
small spacings/
small objects/
fine details

|||
← d → a model for
large spacings in object/
big objects / 'coarse' details

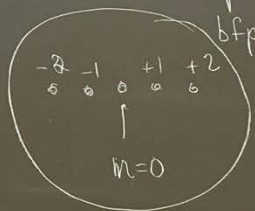
- 2) Abbe observes image in image plane of objective & diffraction pattern in back focal plane (bfp)
ffp = front focal plane
opposite to incident light direction
-

- 3) Image with resolved object details occurs only if at least one ^{higher} diffraction order (e.g. $m = \pm 1$) is captured by objective and appears in bfp.

Passing only $m=0$ gives an unresolved image.

Center spot ($m=0$) reflects undiffracted light
more central spots (e.g. blue) encode bigger details
more peripheral spots (e.g. pink) encode fine details

- 4) Capturing more diffraction spots improves resolution.



- 5) Derived an expression for the smallest resolvable spacing, d_{min} ,
Want small d_{min}
 d_{min} gets smaller (resolution improves) as $\lambda \downarrow$ and $NA \uparrow$
- $$d_{min} = \frac{0.5 \lambda_0}{NA} \quad \text{Abbe} \quad \frac{0.61 \lambda_0}{NA} \quad \text{Rayleigh}$$

Selected Blackboard Images

Lecture 8 (continued)

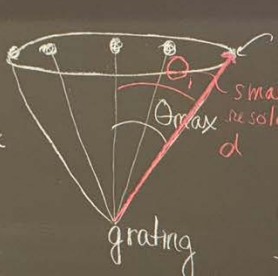
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(6) Abbe Theory
Filtering the diffraction alters the image.
determines

Derivation of Abbe Resolution (to within a factor of 2)

Derivation of Abbe to within a factor of 2

λ_0/n refractive index of medium between sample & lens
 $\lambda_0/n = d \sin \theta_1$
 $\lambda_0/n = d \sin \theta_{\max}$
 $d_{\min} = \frac{\lambda_0}{n \sin \theta_{\max}} = \frac{\lambda_0}{NA}$
 Really $d_{\min} = \frac{\lambda_0}{2NA}$



diffraction spots at angles θ given by $m\lambda = d \sin \theta$
 as $d \downarrow$ angles $\uparrow$ and at some point d is so small that objective just captures θ_1 (first non-zero spot)
 this is $d_{\min}$ when $\theta_1 = \theta_{\max}$

θ_1 smallest resolvable d